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Temporal Difference Learning and Dopaminergic Mechanisms in Bellman-Attention Transformers

Dyuti Ghosh, Dibya Ghosh, Debi Prasad Ghosh debiprasadghosh@gmail.com

Abstract: Transformer architectures have achieved remarkable success in sequence modeling, yet they typically lack explicit mechanisms for integrating reward-based learning signals—particularly those akin to dopaminergic reward prediction errors found in the brain. This conceptual paper proposes **Bellman-Attention**, a novel framework that embeds temporal difference (TD) learning and dopamine-like modulation into the Transformer's self-attention mechanism. By introducing *reward heads* and *future value heads*, Bellman-Attention captures both immediate and discounted returns, enabling the model to prioritize reward-critical information in complex sequential tasks. A dopaminergic gating component further adapts attention weights based on TD errors, mirroring how real-world dopamine bursts and dips reinforce or suppress certain neural pathways. Although no empirical results are provided, we outline potential numerical experiments—ranging from classic control environments to delayed- and sparse-reward tasks—to assess the architecture's performance. We hypothesize that Bellman-Attention would converge faster and handle long-range dependencies more effectively than standard Transformers and purely RL-based methods. Beyond its immediate machine learning applications, the framework presents opportunities for **computational psychiatry**, where disruptions in Bellman-Attention's dopaminergic pathways could conceptually model disorders such as Parkinson's disease, schizophrenia, and addiction. By bridging neuroscience and deep learning, Bellman-Attention offers a promising roadmap for biologically inspired AI research.

1. Introduction

1.1 Motivation

Over the last decade, **Transformer architectures** have become the go-to solution for a variety of sequence modeling tasks, achieving state-of-the-art performance in domains like natural language processing and computer vision (Devlin, Chang, Lee, & Toutanova, 2018; Radford, Narasimhan, Salimans, & Sutskever, 2018). Their ability to capture long-range dependencies and process sequences in parallel has spurred remarkable progress. Despite these successes, Transformers lack intrinsic mechanisms for adapting to **reward-based feedback**, a critical component of real-world decision-making. In contrast, **biological systems**—particularly those governed by **dopaminergic** signaling—excel at using reward prediction errors to guide behavior (Schultz, Dayan, & Montague, 1997). Traditional Transformer attention focuses on statistical correlations in input data but does not prioritize states or tokens associated with potential future rewards. As a result, these models may struggle in **reinforcement learning (RL)** contexts that demand continual adaptation based on evolving reward signals.

1.2 Key Insight

The central premise of our **Bellman-Attention** framework is that self-attention mechanisms can be extended to account for **both immediate and long-term rewards**. By embedding the **Bellman Optimality Equation** (Sutton & Barto, 2018) into the attention computation, we aim to endow the

model with the ability to weigh future returns alongside present signals—much like the brain's dopaminergic pathways do for reward-based learning (Schultz, 2016). Specifically, **temporal difference (TD) errors** act as a gating mechanism that dynamically amplifies or suppresses attention to certain tokens, mirroring how dopamine surges reinforce unexpectedly positive outcomes while dips reduce focus on suboptimal states. This biological parallel enables Bellman-Attention to capture the recursive and predictive nature of RL within the Transformer's parallelizable architecture.

1.3 Scope

This paper is **conceptual**: we do not present empirical validation but rather propose a theoretical design for integrating dopamine-like reward signaling into Transformers. We detail how **reward heads** and **future value heads** could be incorporated to track immediate and discounted rewards, discuss the dopaminergic gating process, and outline **potential numerical experiments** (e.g., CartPole, MountainCar, sparse-reward tasks) to evaluate performance in future work. Additionally, we explore how **Bellman-Attention** might help illuminate **psychiatric disorders**, where aberrations in dopaminergic function contribute to pathologies like Parkinson's, schizophrenia, or addiction (Friston, Stephan, Montague, & Dolan, 2008; Wise, 2004). By linking biologically inspired RL signals with the power of Transformers, Bellman-Attention opens up new directions for both AI research—particularly in **sequential decision-making**—and computational psychiatry, where dysregulations of reward learning can be simulated and studied in a controlled setting. This framework thus offers a promising avenue for creating **adaptive, reward-sensitive** Transformer architectures that reflect the interplay of cognition and neural reward processes.

2. Background

2.1 Transformers and Attention

Transformer architectures have revolutionized sequence modeling by enabling parallelized processing of input tokens and capturing long-range dependencies through self-attention (Devlin, Chang, Lee, & Toutanova, 2018; Radford, Narasimhan, Salimans, & Sutskever, 2018). In the Transformer framework, each token in the input sequence is transformed into a query, key, and value representation. The self-attention mechanism then computes attention scores to identify which tokens in the sequence are most relevant to each other, aggregating those that are highly weighted into the output representation. This approach eliminates the need for recurrent operations and allows for more flexible modeling of context across the sequence.

Despite the notable success of Transformers in natural language processing, computer vision, and other domains, they remain primarily optimized for static sequence-to-sequence or classification tasks. When confronted with **dynamic, reward-based environments**, Transformers lack intrinsic mechanisms for adapting attention weights based on reward signals or learned value functions. Consequently, they may not fully exploit feedback loops that are critical to reinforcement learning, where actions and outcomes evolve and must be integrated into the model's parameter updates.

2.2 Reinforcement Learning Fundamentals

Reinforcement learning (RL) focuses on how agents can learn optimal behaviors through trial-and-error interactions with an environment (Sutton & Barto, 2018). At its core are concepts such as:

- **Value Functions**: A state's value is defined as the expected cumulative reward an agent can obtain starting from that state and following a particular policy.

- **Bellman Optimality Equation**: The Bellman equation provides a recursive formulation for value functions by relating a state's value to the immediate reward plus the discounted value of the next state. Formally,

$$V^*(s) = \max_a \left[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, V^*(s') \right],$$

  where $R(s,a)$ is the reward for taking action a in state s, γ is the discount factor, and $P(s' \mid s,a)$ is the transition probability.

- **Temporal Difference (TD) Learning**: TD methods, such as Q-learning or SARSA, update value estimates based on the difference between observed rewards and current value predictions (the TD error). This mechanism allows an agent to learn from partially complete episodes without waiting for a final outcome.

These principles endow RL agents with the capability to iteratively refine their estimates of long-term rewards, guiding action selection in uncertain and dynamic environments. However, standard Transformers do not natively implement these iterative, reward-focused mechanisms.

2.3 Dopamine's Role in RL

Biological studies underscore the importance of **dopaminergic** systems in learning and decision-making (Schultz, Dayan, & Montague, 1997; Wise, 2004; Schultz, 2016). Dopamine neurons, particularly in the ventral tegmental area and striatum, appear to encode **reward prediction errors**—signals that reflect the difference between expected and actual outcomes. A positive dopamine "burst" suggests a better-than-expected outcome, while a decrease signals a worse-than-expected one. These reward prediction errors drive learning by reinforcing neural pathways associated with rewarding actions and discouraging those that lead to negative or lesser rewards.

Translating these findings into artificial systems has led to the exploration of algorithms that mimic dopaminergic pathways, most notably through temporal difference errors. When such biologically motivated mechanisms are integrated into RL, models can learn not only from actual outcomes but also from the discrepancy between what was predicted and what actually occurred—fostering more robust, sample-efficient learning. Consequently, **embedding dopaminergic principles into Transformer-based architectures** offers a pathway to create systems that adaptively modulate attention in response to reward-relevant feedback, potentially achieving more efficient and human-like decision-making.

3. Proposed Bellman-Attention Architecture

The **Bellman-Attention** architecture extends the standard Transformer's self-attention mechanism by incorporating explicit representations of reward and value, along with a dopaminergic gating process informed by temporal difference (TD) errors. This design aims to endow the model with the ability to prioritize reward-critical information, much like biological systems that reinforce or suppress specific neural pathways based on dopaminergic feedback (Schultz, Dayan, & Montague, 1997; Wise, 2004).

3.1 Core Components

1. **Reward Heads**

- **Motivation**: Biological structures like the orbitofrontal cortex (OFC) estimate immediate rewards and guide moment-by-moment decision-making (Rolls, 2004).
- **Implementation Concept**:
  - Each Transformer layer can include a learnable "reward head" that outputs a scalar or matrix R representing the predicted immediate reward for each token or pair of tokens.
  - When integrated into attention, R directly influences how the model weighs different parts of the input sequence, favoring those associated with higher short-term gains.

2. **Future Value Heads**
   - **Motivation**: Long-range planning in the brain involves dopamine-sensitive pathways that evaluate expected future outcomes (Miller & Cohen, 2001; Schultz, 2016).
   - **Implementation Concept**:
     - A "value head" estimates the discounted sum of future rewards, producing V for each token or token pair.
     - By incorporating γV (where γ is the discount factor) into attention logits, Bellman-Attention captures both immediate and long-term consequences of each decision.
3. **Temporal Discounting**
   - **Motivation**: Humans and animals tend to devalue rewards that occur further in the future (Friston, Stephan, Montague, & Dolan, 2008).
   - **Implementation Concept**:
     - A discount factor γ ∈ [0,1] modulates how strongly future returns contribute to the attention computation.
     - Lower γ emphasizes near-term rewards, while higher γ encourages forward-looking strategies—akin to balancing impulsivity versus deliberation.
4. **Dopaminergic Gating**
   - **Motivation**: Dopamine neurons emit bursts or dips in response to unexpected rewards (positive or negative), thus reinforcing or attenuating synaptic connections (Schultz, Dayan, & Montague, 1997).
   - **Implementation Concept**:
     - Bellman-Attention applies a gating function informed by the TD error $\delta_t = R_{t+1} + \gamma V(s_{t+1}) - V(s_t)$.
     - When δ_t is positive, the gating mechanism amplifies the reward and value signals for the associated tokens in the attention layer; when δ_t is negative, it reduces their influence.
     - This process captures how biological systems dynamically regulate focus based on discrepancies between predicted and actual rewards.

3.2 Mathematical Formulation

1. **Augmented Attention Logits**

   In a standard Transformer, the self-attention logits are:

   $$QK^T \in \mathbb{R}^{B \times H \times L \times L},$$

   where Q and K are projections of the input hidden states (batch size B, number of heads H, sequence length L). Bellman-Attention adds the reward (R) and discounted future value (γV) to these logits, creating:

   $$\alpha_{\text{logits}} = \frac{QK^T + R + \gamma V}{\sqrt{d_k}},$$

   where d_k is the dimensionality of the queries and keys.

2. **SoftMax and Value Combination**

   After computing the attention weights via softmax:

   $$\alpha = \text{softmax}(\alpha_{\text{logits}}),$$

   these weights multiply the value vectors V (from the standard Transformer attention), yielding:

   $$\text{Bellman-Attention}(Q, K, V) = \alpha \times V.$$

3. **Temporal Difference Error Integration**

   - **Definition**: The TD error δ_t compares the next-step value plus reward to the current estimate:

     $$\delta_t = R_{t+1} + \gamma V(s_{t+1}) - V(s_t).$$

   - **Gating Mechanism**: Dopaminergic gating uses δ_t to scale the reward and value terms. For instance, one can update a gating parameter β per layer:

     $$\beta \leftarrow \beta + \eta\, \delta_t,$$

     where η is a learning rate. A positive δ_t increases β, amplifying reward and value signals, while a negative δ_t decreases it.

4. **Parameter Updates**

The network parameters—including those of the reward heads (θ_R), future value heads (θ_V), and gating mechanism (θ_G)—are optimized via backpropagation. A typical loss function could combine a task-specific objective (e.g., cross-entropy or policy gradient) with a penalty on TD error discrepancies:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \sum_t (\delta_t)^2,$$

ensuring that the model refines its reward and value predictions in accordance with observed outcomes.

3.3 Conceptual Advantages

1. **Reward-Driven Focus**: By weaving reward predictions into attention, Bellman-Attention biases the Transformer's receptive field toward reward-critical tokens.
2. **Long-Term Planning**: Future value heads enable the model to account for distant outcomes, a marked shift from the predominantly short-horizon perspective of most Transformers.
3. **Biological Resonance**: Dopaminergic gating echoes the function of dopamine in real neural circuits, potentially offering new insights into both AI-driven reinforcement learning and computational neuroscience (Schultz, 2016).

In summary, **Bellman-Attention** augments the Transformer's self-attention with biologically inspired RL concepts, hypothesizing a pathway to more adaptable, reward-sensitive sequence models. While the design remains conceptual, the next sections propose how one might empirically test its performance and consider its implications for computational psychiatry.

4. Proposed Numerical Experiments

Although no empirical results are presented in this paper, we outline several **numerical experiments** that could be conducted to assess the efficacy of **Bellman-Attention**. These experiments are designed to test key aspects of the proposed architecture, such as reward prediction, future value estimation, dopaminergic gating, and temporal discounting.

4.1 Experimental Scenarios

1. **Discrete Action Tasks (e.g., CartPole)**
    - **Rationale**: CartPole is a classic control task in which an agent must learn to balance a pole on a moving cart, offering immediate feedback and relatively straightforward state transitions (Sutton & Barto, 2018).
    - **Hypothesis**:
        - *Faster Learning*: With dopaminergic gating highlighting reward-critical states, Bellman-Attention is expected to converge more rapidly than a standard Transformer-based policy.
        - *Robustness*: Emphasis on reward heads and value heads should lead to more stable policies, reducing variance in performance over training runs.

2. **Delayed Reward Tasks (e.g., MountainCar, Multi-Step Maze)**
    - **Rationale**: Tasks like MountainCar require agents to accumulate momentum and often receive sparse or delayed rewards. Multi-step mazes introduce path dependencies where each choice impacts distant future states.
    - **Hypothesis**:
        - *Enhanced Performance with Temporal Discounting*: By explicitly modeling discounted future rewards, Bellman-Attention should discover strategies that account for long-term outcomes more effectively than baseline models.
        - *Value Bootstrapping*: Future value heads should guide the agent to promising trajectories, even when intermediate rewards are absent or minimal.
3. **Sparse Reward Environments**
    - **Rationale**: In scenarios where rewards are infrequent—such as exploration-heavy or partially observable tasks—agents often struggle to identify which states are truly relevant for eventual success.
    - **Hypothesis**:
        - *Reward-Critical Focus*: The dopaminergic gating mechanism will amplify attention weights for rare but important reward signals, thereby improving exploration efficiency.
        - *Comparative Advantage*: Standard Transformers without reward-driven modulation may fail to prioritize these sparse but crucial cues, leading to slower or incomplete learning.

4.2 Evaluation Metrics

1. **Episode Reward Sums**
    - **Description**: Track the cumulative reward attained by the agent per episode. Higher totals indicate better task performance.
    - **Relevance**: Provides a direct measure of how effectively Bellman-Attention optimizes rewards across training episodes.
2. **Policy Stability**
    - **Description**: Examine the variability of the agent's actions over time. A convergent policy that reliably balances exploration and exploitation suggests robust learning.
    - **Relevance**: Bellman-Attention's dopaminergic gating is intended to smooth out learning by reinforcing successful trajectories, leading to more stable policy updates.
3. **Alignment with Biological Data**
    - **Description**: Where possible, compare changes in the agent's estimated value functions and TD errors with known patterns of dopaminergic firing in neuroscience studies (Schultz, Dayan, & Montague, 1997; Wise, 2004).

- **Relevance**: Demonstrates whether Bellman-Attention qualitatively replicates dopaminergic reward prediction errors, supporting claims of biological plausibility.

4.3 Expected Outcomes

1. **Faster Convergence and Higher Cumulative Rewards**
    - **Reasoning**: By integrating reward heads, future value heads, and dopaminergic gating directly into the Transformer's attention mechanism, agents are expected to identify high-value actions more swiftly than either standard Transformers or purely RL-based models.
    - **Implication**: If validated, this would indicate that Bellman-Attention leverages biologically inspired RL signals to achieve more sample-efficient learning.
2. **Improved Handling of Long-Range Dependencies**
    - **Reasoning**: Bellman-Attention's explicit modeling of future rewards and temporal discounting should help the agent account for outcomes that occur several steps ahead.
    - **Implication**: Success on tasks featuring long time horizons or sparse rewards would demonstrate the benefit of unifying self-attention with RL-based value estimation principles.

By designing experiments that stress both the **short-term** and **long-term** reward prediction capabilities of Bellman-Attention, researchers could empirically test whether these conceptual innovations align with—or even surpass—the performance of existing RL methods and standard Transformer architectures.

5. Discussion of Expected Results and Biological Alignment

5.1 Biological Plausibility

By integrating temporal difference (TD) errors into the Transformer's self-attention, Bellman-Attention aims to replicate the neuromodulatory role of dopamine in the brain (Schultz, Dayan, & Montague, 1997; Wise, 2004). In biological systems, dopaminergic neurons encode reward prediction errors, amplifying neural responses to unexpectedly high rewards and attenuating responses when outcomes are worse than expected (Schultz, 2016). A similar mechanism in Bellman-Attention—via *dopaminergic gating*—highlights reward-critical information and suppresses attention to less promising states. As a result, we would expect learning patterns that mimic core aspects of reward-based plasticity in the brain, including rapid adaptation to novel stimuli and reinforcement of high-value trajectories.

Moreover, integrating future value estimates aligns with how the prefrontal cortex and basal ganglia evaluate and plan for long-range outcomes (Miller & Cohen, 2001). By explicitly representing discounted future rewards, Bellman-Attention could capture delayed benefits in its attention computations, mirroring the neural substrates that help humans and animals consider distant goals. If the proposed framework performs as anticipated, it would demonstrate a step toward *human-like* decision-making in AI, deepening the parallel between reinforcement learning models and actual neural processes.

5.2 Potential Limitations

While Bellman-Attention incorporates biologically inspired elements, it also involves **oversimplifications** of dopaminergic function. In reality, dopamine interacts with other neuromodulators—such as serotonin and norepinephrine—that further refine learning, mood regulation, and attention (Rolls, 2004). By modeling dopamine in isolation, the proposed architecture may not fully capture the complexity of neuromodulatory interactions observed in the brain. Additionally, the straightforward mapping of a temporal difference error to "dopaminergic gating" abstracts away the spatial and temporal variability of actual neural firing patterns.

In practical terms, Bellman-Attention introduces **greater computational overhead** compared to standard Transformers. Reward heads, future value heads, and the gating mechanism each add to the parameter count and the number of operations required per training step. While these additions may pay off in improved learning efficiency, they could make the model less tractable for tasks demanding very large-scale architectures or real-time inference. Future optimizations—such as pruning or sharing parameters among heads—may help mitigate these costs.

5.3 Extensions and Future Research

An important direction for improving biological fidelity involves **incorporating additional neuromodulators**. For instance, serotonin has been implicated in temporal stability and aversion processing, while norepinephrine relates to arousal and attention-gating (Friston, Stephan, Montague, & Dolan, 2008). Extending Bellman-Attention to include multiple modulatory signals could enable a more nuanced simulation of brain function, potentially enhancing the model's adaptability in complex or uncertain environments.

Another promising avenue is to **make discount factors and gating parameters adaptive**. In real brains, neuromodulatory systems continually update their sensitivity and baseline firing rates depending on an individual's physiological state or environmental contingencies (Schultz, 2016). Allowing Bellman-Attention to dynamically adjust γ or the dopaminergic gating coefficient based on recent performance or volatility in the task could yield even more robust and flexible behaviors. This would mirror how living organisms fine-tune their pursuit of short-term versus long-term rewards in response to changing conditions.

6. Computational Psychiatry Implications

6.1 Modeling Disorders

The **Bellman-Attention** framework, by design, ties reward processing directly to a dopaminergic-like signal. This opens avenues for conceptualizing how disruptions in dopamine function might lead to psychiatric or neurological disorders (Friston, Stephan, Montague, & Dolan, 2008; Schultz, 2016):

- **Parkinson's Disease**: Reduced dopamine levels in Parkinson's can be approximated by lowering the amplitude of the dopaminergic gating or diminishing the weight of temporal difference errors. In doing so, the model would devalue potential future rewards, mirroring the cognitive and motor symptoms arising from dopamine depletion in the basal ganglia.
- **Schizophrenia:** Hyperactive dopamine signaling in certain pathways has been implicated in schizophrenia. Within Bellman-Attention, an exaggerated gating response to TD errors could lead to over-amplification of certain states or actions, producing erratic attention shifts or

hallucination-like behaviors. Such a model might help theorize why patients with schizophrenia assign excessive salience to irrelevant stimuli (Rolls, 2004).

- **Addiction:** Substance use disorders often involve an overemphasis on immediate rewards and a neglect of long-term consequences. By reducing the effective discount factor (γ) and increasing dopaminergic gating sensitivity for short-term rewards, Bellman-Attention might demonstrate compulsive action selection. This parallels how drugs of abuse often hijack dopaminergic circuits, reinforcing reward-seeking behavior at the expense of future well-being (Wise, 2004).

These simulations remain conceptual, but they illustrate how modifying **dopaminergic parameters** or reward/value components in Bellman-Attention could replicate fundamental aspects of psychiatric and neurological conditions.

6.2 Future Directions in Clinical Research

One of the most promising applications of Bellman-Attention lies in its potential as a **computational testbed** for exploring novel interventions:

- **Targeted Interventions:** By systematically altering dopaminergic gating coefficients or the discount factor, researchers could simulate how pharmaceutical agents (e.g., L-DOPA, antipsychotics) might restore or modulate reward signals in the model. Observing changes in the agent's policy or attention patterns could yield insights into the potential efficacy of different drug regimens.

- **Disease Progression Modeling:** Clinical conditions often unfold gradually, with dopaminergic function deteriorating or becoming dysregulated over time (e.g., Parkinson's progressing from mild tremors to severe motor impairment). Adjusting Bellman-Attention's parameters to degrade incrementally could provide a simplified, yet illustrative view of disease trajectories, informing long-term treatment strategies.

Through these future-oriented approaches, Bellman-Attention may help bridge the gap between **reinforcement learning research** and **clinical neuroscience**, offering a platform to test hypotheses about the neurobiological underpinnings of mental disorders and prototype interventions in a controlled, computational setting.

7. Conclusion

7.1 Summary

In this conceptual paper, we have presented **Bellman-Attention**, a framework designed to embed reinforcement learning (RL) principles and dopaminergic-like signals into the Transformer architecture. By incorporating *reward heads, future value heads, temporal discounting,* and a *dopaminergic gating* mechanism, Bellman-Attention aspires to unify the powerful sequence modeling capabilities of Transformers (Devlin, Chang, Lee, & Toutanova, 2018; Radford, Narasimhan, Salimans, & Sutskever, 2018) with core insights from biological RL (Schultz, Dayan, & Montague, 1997; Sutton & Barto, 2018). In doing so, it holds the potential to manage both immediate and long-term rewards more effectively, reflecting the way dopamine modulates attention and learning in the human brain.

7.2 Relevance

The conceptual contributions of Bellman-Attention are twofold. First, it provides an **architectural blueprint** for next-generation AI systems that can adaptively weight reward-critical information, potentially improving efficiency and performance in complex decision-making tasks. Second, by mimicking dopaminergic reward prediction errors, it offers a **computational window** into psychiatric and neurological phenomena. Through controlled manipulation of dopaminergic-like parameters, Bellman-Attention could illuminate the mechanisms underlying conditions such as Parkinson's disease, schizophrenia, and addiction (Wise, 2004; Friston, Stephan, Montague, & Dolan, 2008). This dual relevance underscores the model's potential to inspire new research at the intersection of cognitive neuroscience, machine learning, and computational psychiatry.

7.3 Open Questions

Despite its promise, Bellman-Attention remains at a **theoretical stage**. Empirical validation—spanning classic RL benchmarks, delayed reward tasks, and real-world scenarios—will be critical for testing the framework's viability and scalability. Additional research is also warranted into the **ethical and clinical implications** of using biologically inspired AI systems, especially if they can emulate or predict mental health conditions. Issues such as data privacy, consent, and the responsible deployment of AI-driven interventions must be carefully addressed. Furthermore, future studies might incorporate multiple neuromodulators, adaptive gating parameters, and larger-scale experiments to fully realize the architecture's potential. By tackling these open questions, Bellman-Attention may serve as both a **powerful AI paradigm** and a **computational lens** for understanding the rich tapestry of cognition and behavior.

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Appendix A: Bellman-Attention

Below is a concise walkthrough of how the Bellman-Attention terms can be arranged to preserve proper shapes and enable valid tensor operations. We will focus on the crucial step:

$$\text{Augmented Attention}(Q, K, V) \;=\; \text{softmax}\Big(\frac{QK^T + R + \gamma\, V_{(\text{for attention logits})}}{\sqrt{d_k}}\Big)\, V,$$

and verify the dimensional consistency of all components involved.

A.1. Standard Transformer Dimensions

Let:

- B = batch size
- H = number of attention heads
- L = sequence length
- d_k = dimensionality of the queries/keys
- d_v = dimensionality of the values

In a multi-head self-attention module, the shapes of the standard tensors are:

1. **Queries** (Q): $[B, H, L, d_k]$
2. **Keys** (K): $[B, H, L, d_k]$
3. **Values** (V): $[B, H, L, d_v]$

When we compute QK^T, the result is:

$$QK^T : [B, H, L, L].$$

This $[B, H, L, L]$ tensor is typically passed through a softmax along the last dimension before being multiplied by V.

A.2. Augmenting With Reward and Value Terms

2.1 Reward Term (R)

- **Goal**: Inject an immediate reward signal into the attention logits.
- **Shape Requirement**: To add R elementwise to QK^T, R must have the same shape: $[B, H, L, L]$.

How to achieve $[B, H, L, L]$?

- One approach is to have a *reward head* produce a scalar reward $\hat{r}_i$ for each token i. You can then broadcast these scalars to form a matrix $[B, H, L, L]$ by combining row-wise and column-wise reward signals (e.g., $\hat{r}_{\text{query}} + \hat{r}_{\text{key}}$).

- Alternatively, the reward head can directly generate a matrix of pairwise reward contributions, effectively capturing how each query token aligns with each key token with respect to immediate reward.

2.2 Future Value Term (γ V)

- **Goal**: Incorporate the discounted future return signal into the attention logits.
- **Shape Requirement**: Similarly, the term γV must be $[B, H, L, L]$ if it is to be added directly to QK^T.

How to achieve $[B, H, L, L]$?

- A *future value head* often produces a single scalar $\hat{v}_i$ per token i for the next-step (or multi-step) discounted return. As with the reward term, you can construct a full $[L, L]$ matrix via a broadcasting approach or by explicitly modeling pairwise future value signals.
- Once shaped to $[B, H, L, L]$, it can be scaled by γ and added to QK^T.

Hence, after ensuring both R and γV are shaped to $[B, H, L, L]$, we can sum them with QK^T:

$$QK^T \in [B, H, L, L], \quad R \in [B, H, L, L], \quad (\gamma V) \in [B, H, L, L].$$

This sum yields $[B, H, L, L]$, which is the correct shape for the softmax operation along the last dimension.

A.3. Final Attention Multiplication With VVV

After computing and normalizing the augmented logits via SoftMax, we obtain a tensor:

$$\alpha = \text{softmax}\left(\frac{QK^T + R + \gamma\, V_{(\text{logits})}}{\sqrt{d_k}}\right),$$

where $\alpha \in [B, H, L, L]$.

The final step is to multiply α by the **value** tensor V, which has shape $[B, H, L, d_v]$. This multiplication follows the standard attention pattern:

$$\text{Bellman-Attention}(Q, K, V) = \alpha \times V,$$

resulting in an output of shape:

$$[B, H, L, d_v].$$

Depending on the implementation, you might then combine (e.g., concatenate and project) across the heads to get a final output of shape $[B, L, d_{\text{model}}]$, consistent with typical Transformer layers.

A.4. Summary of Dimensional Consistency

1. $\mathbf{QK^T}$: $[B, H, L, L]$
2. $\mathbf{R}$: $[B, H, L, L]$ after appropriate broadcasting or pairwise modeling.
3. $\gamma\mathbf{V}$ (for attention logits): $[B, H, L, L]$ after shaping to match (QK^T).
4. **Elementwise addition**:

$$(QK^T + R + \gamma V) \in [B, H, L, L].$$

5. **Softmax** across last dimension: $\alpha \in [B, H, L, L]$.
6. **Final multiplication with $\mathbf{V}$ (values)**:

$$\alpha \times V \;\in\; [B, H, L, d_v].$$

By carefully ensuring R and γV are generated (or broadcast) to the same $[B, H, L, L]$ shape as QK^T, **Bellman-Attention** maintains the standard Transformer attention flow while adding reward-driven and value-driven modulation. This design preserves dimensional consistency throughout the forward pass.

Appendix B: Bellman-Attention to Transformer Stack

Below is a conceptual overview of how Bellman-Attention integrates into a multi-layer Transformer stack, ensuring that both the **reward** and **value** signals, as well as **dopaminergic gating**, propagate through each layer:

B.1. Layer-by-Layer Forward Pass

1. **Input to Layer ℓ**

   - Let $\mathbf{h}^{(\ell-1)} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$ represent the hidden states passed from the previous layer (or the initial input embeddings if $\ell = 1$).

   - These hidden states are first projected into **queries**, **keys**, and **values** for the multi-head attention mechanism.

2. **Compute Reward and Value Signals**

   - **Reward Heads**: Produce a matrix $R^{(\ell)}$ reflecting immediate reward estimates or relevance for each pair of tokens. Conceptually, the head might map $\mathbf{h}^{(\ell-1)}$ to a shape that can be broadcast to $[B, H, L, L]$.
   - **Future Value Heads**: Output a matrix $V^{(\ell)}$ of future value estimations, also shaped (via broadcasting or pairwise modeling) to $[B, H, L, L]$.

   - These signals get incorporated into the attention *logits* (before SoftMax) so that each token's attention to every other token is modulated by both **reward** and **future value**.

3. **Apply Dopaminergic Gating**

   - Based on the **temporal difference (TD) error** δ_t (which can be computed at a higher level or per environment step if this model is unrolled in an RL scenario), the gating mechanism adjusts how strongly $R^{(\ell)}$ and $V^{(\ell)}$ influence the attention scores.
   - This mimics dopamine bursts (positive δ_t) increasing attention to reward-relevant tokens, and dopamine dips (negative δ_t) reducing it.

4. **Augmented Attention Computation**

   - The core Bellman-Attention operation at layer ℓ is:

     $$\alpha^{(\ell)} = \text{softmax}\left(\frac{QK^T + R^{(\ell)} + \gamma V^{(\ell)}}{\sqrt{d_k}}\right),$$

     where γ is the discount factor, and d_k is the dimensionality of the keys.
   - Multiply $\alpha^{(\ell)}$ by the value matrix (in the standard sense of Transformer attention) to get the final attended representation:

     $$\mathbf{z}^{(\ell)} = \alpha^{(\ell)} \times \text{Values}.$$

5. **Feed-Forward and Residual Connection**

- As in a standard Transformer, the output $\mathbf{z}^{(\ell)}$ then goes through:
  1. A feed-forward sub-layer (often a position-wise MLP).
  2. A residual connection and layer normalization.
- The resulting hidden states $\mathbf{h}^{(\ell)}$ become the input to the next layer, $\ell + 1$.

B.2. Higher-Level Reinforcement Learning Loop

- In a typical RL setup, the agent (Bellman-Attention Transformer) receives **state observations** (possibly tokenized). The output at the final layer, $\mathbf{h}^{(L)}$, can be used:
  1. To **select actions** (if this is a policy network or a Q-network variant).
  2. To **update** the reward and value predictions for the next step.
- **Temporal Difference Error** (δ_t) is computed at the end of each RL step (or episode), then **backpropagated** through the network. This means that **reward heads, value heads,** and **dopaminergic gating parameters** each receive gradient updates that train them to better predict and weigh reward-related signals.

B.3. Parameter Updates and Loss Computation

1. **Bellman-Attention Loss**
   - In addition to any task-specific or policy-related loss (e.g., a Q-learning loss or a policy gradient objective), Bellman-Attention incorporates a **TD-error** term. For example:

$$\mathcal{L}_{\text{Bellman}} = \mathcal{L}_{\text{task}} + \lambda \sum_t (\delta_t)^2.$$

   - Here, δ_t is the RL prediction error, which also drives the **dopaminergic gating** adjustments.
2. **Backpropagation Through Layers**
   - **Gradients** flow from the final layer L back to layer 1, updating:
     1. **Reward heads** (θ_R): to better predict immediate rewards.
     2. **Value heads** (θ_V): to reduce discrepancy between predicted and actual future returns.
     3. **Gating parameters** (θ_G): to refine how TD errors scale the attention logits.
   - Because each layer has its own attention block, each block learns how to optimally incorporate reward and value signals at progressively **deeper** representations.

B.4. Summary of Layer Propagation

- **Each Layer** in Bellman-Attention has **its own** reward heads, value heads, and a gating mechanism.
- After computing standard queries, keys, and values, the reward and future value signals are broadcast or shaped to match the $[B, H, L, L]$ attention logits.
- **Dopaminergic gating** then applies a scalar or matrix-based modulation informed by δ_t.
- The **augmented logits** go through a SoftMax, multiply by the value vectors, and proceed to a feed-forward sub-layer.
- The **output** of each layer ℓ contains updated representations that reflect both short-term and long-term reward considerations, passed on to layer $\ell + 1$.
- Ultimately, the final layer's output is used for **action selection** or **prediction**, and the entire architecture is trained end-to-end via RL losses that include the TD-error component.

In this way, Bellman-Attention's reward-driven modifications **propagate** from the first layer to the last, continually refining how the Transformer attends to different parts of the input in response to reward signals—much like a multi-layer neural analog of dopaminergic feedback in biological systems.

Appendix C: Parallelization Capacity

One of the key strengths of Transformer architectures is their ability to process input tokens **in parallel** rather than sequentially, thus achieving efficient training and inference, especially on modern GPU or TPU hardware (Devlin et al., 2019; Radford et al., 2018). **Bellman-Attention** preserves this parallelization advantage by making only minimal modifications to the attention mechanism:

1. **Broadcasting Reward and Value Signals**
    - In Bellman-Attention, the reward (R) and discounted future value (γV) matrices are shaped or broadcast to match the dimensions of the attention logits $[B, H, L, L]$.
    - This broadcasting step can be performed in parallel for each head and each position in the sequence, requiring no sequential operations.

2. **Maintaining the $O(L^2)$ Attention Pattern**
   - Like standard Transformers, Bellman-Attention computes self-attention by generating a $[B, H, L, L]$ matrix for each head, which is then normalized via softmax and multiplied by the $[B, H, L, d_v]$ value tensor.
   - While the addition of R and γV to the attention logits increases the complexity of the logic, it does **not** add extra sequential dependencies. All operations still occur within an $O(L^2)$ attention step that can be parallelized across tokens and heads.
3. **Dopaminergic Gating**
   - The gating function, which scales reward and value terms according to a temporal difference error (δ_t), is also performed elementwise or in a broadcast manner.
   - Whether δ_t is computed once per sequence, per token, or per environment step, the gating multiplications remain parallelizable across the batch and sequence length. There is no inherent need for sequential iteration within a single layer of Bellman-Attention.
4. **Batch and Model Parallelism**
   - Bellman-Attention is fully compatible with existing parallelization strategies for Transformers—such as distributing heads or splitting batches across multiple GPUs or TPUs.
   - The additional heads (reward and future value) can be integrated into standard data-parallel or model-parallel frameworks without significant architectural changes.

By maintaining the core dataflow of Transformer attention, **Bellman-Attention** ensures that all major computations—query-key multiplication, the addition of reward/value, SoftMax normalization, and the final matrix multiplication with values—can be carried out in parallel across the entire sequence. Consequently, **the proposed modifications do not compromise the fundamental parallelization capacity** that makes Transformers so appealing for large-scale sequence modeling tasks.

Appendix D: Key-Value Caching in Bellman-Attention

Key-value (**KV**) caching is a technique used in standard Transformer decoders to **speed up autoregressive inference** by storing previously computed keys (K) and values (V). Instead of recomputing these tensors for every new token, the decoder reuses cached values, allowing the model to process sequences incrementally. Below, we outline how this mechanism can be adapted for **Bellman-Attention**, ensuring that the additional reward- and value-related computations remain compatible with **fast decoding**.

D.1 Standard KV Caching Recap

In a typical Transformer decoder (e.g., for language generation):

1. **Keys and Values** for tokens $1 \ldots t-1$ are computed once, then stored in a cache.
2. When predicting the next token t, only the new query $\mathbf{Q}_t$ is computed. The model then **concatenates** $\mathbf{Q}_t$ with the cached keys and values to produce the attention output.
3. This process repeats for tokens $t+1, t+2, \ldots$, allowing each step to be computed in $O(t)$ rather than $O(t^2)$.

In a

D.2 Extending KV Caching to Bellman-Attention

1. Reward Heads and Future Value Heads

- **Key Idea**: In Bellman-Attention, each layer incorporates *reward heads* (R) and *future value heads* (V) that produce terms shaped to match $[B, H, L, L]$ at training time (full-sequence context).
- **Caching Approach**:
  - During autoregressive decoding, at time step t, the model only has partial context (tokens $1 \ldots t-1$).
  - Similar to how keys and values are cached, the outputs of reward heads R and future value heads V for previous tokens can be stored.
  - For each new token, Bellman-Attention **concatenates** the newly computed reward/value terms with the cached ones, just like standard K and V.
  - This allows the attention mechanism to incorporate previously computed reward/value estimates without recalculating them for all past tokens.

2. Dopaminergic Gating

- **TD Errors in Incremental Mode**:
  - A challenge arises in **autoregressive RL** scenarios: the temporal difference (TD) error δ_t depends on rewards and future value estimates, which may not be fully known at inference time.
  - For tasks where partial environment feedback is available at each time step, the gating coefficient can be updated incrementally or predicted from partial context.

- **Caching Gating Parameters**:
  - The gating coefficient(s) that modulate R and γV can also be tracked across steps.
  - If gating updates ($\beta \leftarrow \beta + \eta\, \delta_t$) are performed on a step-by-step basis, the new dopaminergic gating parameter β simply applies to the attention logits for the current and future tokens, without altering the cached past.

D.3 Computational Efficiency

1. **Attention Logits**
   - By storing the keys, values, and the reward/value signals for each past token, Bellman-Attention avoids recomputing them from scratch.
   - When token t arrives, the model computes:

     $$\alpha_{\text{logits}}^{(t)} = \frac{Q_t K_{\text{cached}}^T + R_{\text{cached}} + \gamma V_{\text{cached}}}{\sqrt{d_k}},$$

     where K_{cached}, R_{cached}, and V_{cached} aggregate information from previous steps.
2. **Parallelization**
   - The same parallel attention mechanism that benefits standard Transformer decoders applies here, as caching does not introduce any new sequential dependencies.
   - Bellman-Attention maintains the O(t) complexity at step t for combining queries with cached data.
3. **Implementation Details**
   - Implementation can mirror that of existing Transformer-decoder frameworks (e.g., huggingface/transformers, Fairseq), with additional buffers for **reward** and **value**.
   - Minimal overhead is introduced by storing these extra tensors, but the mechanism remains otherwise the same: new queries are computed for the new token, then combined with cached items to produce the attention output.

D.4 Summary

Key-value caching in Bellman-Attention follows the same principles as standard Transformer decoder caching, with the extension that **reward** and **future value** terms must also be cached alongside the usual keys and values. The **dopaminergic gating** mechanism, while conceptually linking past and present states via TD errors, does not disrupt the cache-based parallelization: gating parameters can be updated incrementally, and partial environment feedback can be used to adapt the gating in real-time.

By retaining parallelizable operations and caching, **Bellman-Attention** remains **scalable** and **efficient** for autoregressive tasks, much like classic Transformers—yet it embeds richer RL-driven signals that potentially yield more adaptive and reward-focused sequence predictions.

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